

ZAINA ZIA

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SUMMARY

Full-stack developer with hands-on experience building and shipping production web applications across the entire stack — React.js, Next.js, Flask, FastAPI, and PostgreSQL. Proven ability to own features end-to-end, from normalized database schema design to pixel-accurate responsive UI. Experienced with third-party API integration including Spotify OAuth and RESTful services, low-level embedded systems programming in 8051 Assembly, and professional software testing in an AI product environment. Comfortable working across the stack independently and delivering to real-world specifications.

SKILLS

Languages: Python, JavaScript (ES6+), C++, SQL, HTML5, CSS3

Frontend: React.js, Next.js, Responsive Design, Figma-to-Code, Cross-device UI

Backend & Databases: Flask, FastAPI, Node.js, RESTful API Design, PostgreSQL, MySQL, Oracle DB, PL/SQL, Supabase

Auth & Integrations: HTTP cookie-based sessions, Postman, REST endpoint testing

AI/ML & Data: NumPy, Pandas, Data pipelines

Tools: Git, GitHub, VS Code, Vercel, Google Colab, JavaFX

Testing: Manual testing, regression testing, bug documentation, API validation, structured reporting

EDUCATION

Bachelor of Computer Science | NUST — SEECS, Islamabad

Sep 2024 – Present

4th Semester | Relevant Coursework: Data Structures & Algorithms, OOP, Advanced Database Systems (Oracle/PL/SQL), Artificial Intelligence, Web Technologies, Theory of Automata

EXPERIENCE

QA Engineer | CodeKnitters — NSTP, NUST Islamabad

Feb 2026 – June 2026

- Tested and documented 30+ reproducible bugs across mobile app builds; covering ANR crashes, UI localization breakdowns, and core feature failures, with full reproduction steps, severity ratings, and environment context
- Validated AI-powered features including prompt-to-image generation, background segmentation, and face detection; reviewed model outputs for correctness and consistency across builds
- Ran structured regression testing across multiple build versions; maintained a shared bug-tracking sheet with per-build status columns, updated collaboratively with the dev team
- Tested all REST API endpoints via Postman, verified request/response integrity, and flagged data inconsistencies between backend and UI layers

Front-End Developer (React.js) | Freelance — Upwork

Oct 2025 – Dec 2025

- Built responsive UI components from Figma designs using React.js and Next.js; ensured consistent cross-device rendering and pixel-accurate design implementation
- Managed version control via Git/GitHub with clean commit history; delivered all milestones on time per client specifications

Python & CS Tutor | Private — Remote (Abu Dhabi Students)

2024 – Present

- Designed and delivered personalized Python and Computer Science lessons for international students, adapting content and pacing to individual learning needs

PROJECTS

Smart Student Fatigue & Workload Tracker

FastAPI · Oracle DB · HTML/CSS/JS · PL/SQL | Apr 2026

- Built a full-stack three-tier application: Oracle relational database, Python FastAPI middleware, and an HTML/JS frontend; implementing 5 role-based dashboards (Student, Counselor, Faculty, Admin) with HTTP cookie-based session management
- Designed adaptive recommendation engine with streak-based lifecycle, tiered alert escalation, and deduplication logic for automated counselor assignment based on computed Burnout Risk Index

Tic-Tac-Toe on AT89S51 Microcontroller

8051 Assembly · Embedded Hardware | March 2026

- Programmed a fully functional two-player game in pure 8051 Assembly on the AT89S51 microcontroller; implemented win detection across all 8 combinations, software debouncing, and bitmask-driven LED control with no OS or library support
- Built and validated hardware on a breadboard (18 LEDs, 9 push-buttons); simulated in Proteus ISIS and flashed via USBasp ISP programmer

Moodify — AI Music Recommendation App

Next.js · Spotify API · OAuth 2.0 | Nov 2025

- Integrated Spotify Web API with OAuth 2.0 to retrieve and process user listening history; built mood-detection logic to serve personalized track recommendations
- Implemented multilingual UI (English/Urdu) with dynamic mood-based theming; deployed on Vercel with full cross-device responsiveness

Legal Case Management System

Flask · React.js · PostgreSQL · Supabase | Apr 2025

- Built full-stack case management application as 2nd semester project with a RESTful Flask backend, React.js frontend, and PostgreSQL database hosted on Supabase
- Designed normalized database schema for case records, client data, and document tracking; tested all API endpoints via Postman for data integrity across every layer

Galacticos FC Management System

JavaFX · Java · MySQL | Apr 2025

- Developed a desktop application with dual login modules (Admin and Player) with role-specific dashboards and access control; Admin manages rosters, matches, and performance data; Player views schedule and submits post-match ratings

COMPETITIONS & ACHIEVEMENTS

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- **Cadenza Challenge — ICASSP 2025** | *Lyric Intelligibility Prediction using ML (NumPy, Pandas)*; built and evaluated predictive models on audio/speech datasets
 - **Build with AI Hackathon** — developed an AI-based automation prototype
 - **Code-o-Clock (NUST)** — algorithmic problem-solving competition
 - **Elevatech Programming Competition** — logic-based coding challenges